Dataset Details — 5G Network Operations (NetOps) Dataset

Item	Value
Name	5G Network Operations (NetOps) Dataset
Source	Hugging Face — https://huggingface.co/datasets/fenar/5g_netops (file `5G_netops_data_100K.csv.xz`)
Alternative	RANalyzer Dataset — https://github.com/wineslab/RANalyzer-Dataset (not used)
Format	CSV (xz-compressed), loaded with `pandas.read_csv("hf://datasets/fenar/5g_netops/5G_netops_data_100K.csv.xz")`
Size	100,000 records × 27 columns; no duplicates; no missing values
Nature	Synthetic network-operations records (about 38K distinct city names, 59 state/territory codes)
Local copy	`03_Source_Code/data/raw/5G_netops_data_100K.csv` (downloaded automatically by `python -m data.ingest`)
Sample	`06_Dataset_Sample.csv` (first 200 rows plus derived fields)

Key fields

field	type	min	median	max	mean
Cell Availability (%)	numeric	74.34	89.40	99.00	89.25
MTTR (hours)	numeric	0.50	5.25	10.00	5.25
Throughput (Mbps)	numeric	114.99	549.21	1283.86	586.57
Latency (ms)	numeric	4.62	77.22	175.84	79.22
Packet Loss Rate (%)	numeric	0.10	6.24	17.00	6.46
Call Drop Rate (%)	numeric	0.00	0.00	6.90	0.95
Handover Success Rate (%)	numeric	90.00	95.02	100.00	95.02
Alarm Count	numeric	0.00	10.00	26.00	10.01
Critical Alarm Count	numeric	0.00	6.75	13.86	6.36
Parameter Changes	numeric	0.00	10.00	19.00	9.52
Successful Configuration Changes (%)	numeric	90.00	94.49	99.00	94.50
Data Usage (GB)	numeric	10.00	25.16	50.00	26.11
User Count	numeric	100.00	5060.00	9999.00	5055.09
Signal Strength (dBm)	numeric	-120.00	-90.06	-60.00	-90.02
Jitter (ms)	numeric	0.10	25.00	50.00	25.01
Connection Setup Success Rate (%)	numeric	90.00	94.96	99.00	94.93
Security Incidents	numeric	1.00	10.00	19.00	10.00
Authentication Failures	numeric	0.00	8.00	63.00	13.25
Temperature (°C)	numeric	-10.00	17.47	40.00	16.25
Humidity (%)	numeric	20.00	54.90	90.00	55.00
Fault Occurrence Rate (%)	numeric	0.00	39.46	81.72	38.16
Season	categorical		4 values		Winter/Spring/Summer/Fall
Weather	categorical		4 values		Clear/Clouds/Rain/Snow
Issue Reported	categorical		2 values		yes/no flag
City	categorical		38066 values		city name (synthetic)
State	categorical		59 values		US state/territory code
Zip	categorical		63145 values		postal code

Derived fields (created by the ingestion pipeline)

Field	Description
`case_id`	Stable identifier `NOC-000000` … `NOC-099999`
`c_<condition>` (21 columns)	0 = normal, 1 = degraded (worse than fleet P75/P25), 2 = severe (P90/P10)
`conditions`	Comma-separated list of the conditions that are present

Field	Description
`category`, `category_name`	Incident taxonomy class (8 categories + Normal)
`risk_score`	0–100 composite (fault likelihood 50 %, degradation breadth 30 %, customer impact 20 %)
`priority`, `severity`	P1 Critical / P2 Major / P3 Minor / P4 Low
`text`	Incident narrative used for semantic retrieval and LLM grounding

Condition thresholds (computed from the data)

condition	KPI	bad when	degraded at	severe at	share degraded+
high_latency	Latency (ms)	high	105.39	129.92	25.0%
packet_loss	Packet Loss Rate (%)	high	9.33	11.88	25.0%
low_throughput	Throughput (Mbps)	low	394.30	304.25	25.0%
call_drops	Call Drop Rate (%)	high	1.66	3.13	25.0%
handover_failure	Handover Success Rate (%)	low	92.51	91.01	25.0%
poor_signal	Signal Strength (dBm)	low	-105.06	-114.04	25.0%
high_jitter	Jitter (ms)	high	37.50	44.98	25.0%
alarm_storm	Alarm Count	high	12.00	14.00	30.5%
critical_alarms	Critical Alarm Count	high	8.42	9.57	25.0%
low_availability	Cell Availability (%)	low	83.19	79.43	25.0%
setup_failure	Connection Setup Success Rate (%)	low	92.49	90.99	25.1%
config_churn	Parameter Changes	high	15.00	18.00	25.1%
config_failure	Successful Configuration Changes (%)	low	92.24	90.89	25.0%
high_load	User Count	high	7526.00	9002.10	25.0%
high_data_usage	Data Usage (GB)	high	37.59	45.07	25.0%
security_incidents	Security Incidents	high	15.00	18.00	26.2%
auth_failures	Authentication Failures	high	18.00	35.00	26.4%
high_fault_rate	Fault Occurrence Rate (%)	high	48.15	55.51	25.0%
slow_repair	MTTR (hours)	high	7.62	9.06	25.1%
extreme_temperature	Temperature (°C)	both	36.05	39.18	10.0%
high_humidity	Humidity (%)	high	76.96	85.10	10.0%

Distribution of derived labels

category	cases	share
Equipment Fault & Alarms	25548	25.5%
Radio Access & Coverage	14499	14.5%
Transport & QoS Degradation	13672	13.7%
Configuration & Change Management	12321	12.3%
Capacity & Congestion	11800	11.8%
Mobility & Handover	11161	11.2%
Security & Authentication	6583	6.6%
Environmental / Weather Impact	2211	2.2%
Normal Operation	2205	2.2%

priority	cases	share
P1	15682	15.7%
P2	16762	16.8%
P3	27721	27.7%
P4	39835	39.8%

Data-quality findings

- `Issue Reported` is about 50/50 and independent of all KPIs (identical group means), so it is treated as noise and

used only as a filter.

- `Fault Occurrence Rate` is strongly determined by Critical Alarm Count (r = 0.93), Authentication Failures (0.64) and Call Drop Rate (0.46). Throughput is strongly anti-correlated with Critical Alarm Count (r = –0.96).
- Most other KPIs are mutually independent (uniform-like synthetic generation), which limits the achievable accuracy of the performance models (see `model_metrics.json`).
- Season has a mild effect: Winter has a higher fault rate (41.9 % vs about 37 %), and Summer has higher latency.

Example narrative (record NOC-000000)

```
Winter network incident at Schneiderville, MN. Weather clouds, -7°C, humidity 29%. Observed conditions: poor signal strength (-110
```